Data Collection Report: Mobile Money Usage in Cameroon

1. Overview

This project set out to understand how mobile money is actually used in Cameroon, with a focus on MTN MoMo and Orange Money. Rather than relying only on platform-level assumptions, the study collected first-hand transaction records and demographic information from real users across different cities, occupations, and income levels.

The goal was not statistical generalization. With a small sample, the priority was to collect rich, trustworthy, and ethically obtained data that could support exploratory analysis and later modeling. For that reason, purposive sampling was used instead of random sampling.

2. Purpose of the Collection

Mobile money has become a major part of everyday financial life in Cameroon. People use it to send money, receive payments, save temporarily, pay for services, and manage day-to-day transactions. This project aimed to capture that behavior in a way that reflects real diversity in usage patterns.

The collection was designed to answer a simple question: who is using mobile money, how are they using it, and what social and economic factors help explain those patterns?

3. Sampling Strategy

Because the project involved only 10 participants, random sampling would likely have produced a narrow and repetitive group. Instead, purposive sampling was used to deliberately include variation across key dimensions:

- Age
- Gender
- Profession
- Income level
- Household size
- Geographic location

This approach helped ensure that the dataset included users whose behavior might differ meaningfully from one another. The participants came from Yaounde, Bamenda, and Douala, which also helped reduce overreliance on a single local network.

The final participant pool included university students, traders, retirees, and public servants, with ages ranging from 18 to 55+ and household sizes from 1 to 9. Monthly incomes ranged from under 50,000 FCFA to above 400,000 FCFA.

4. Data Collected

Each participant was asked to provide two types of information:

- 10 mobile money transaction records
- A short demographic questionnaire

The transaction records gave insight into real usage behavior, while the questionnaire provided context for interpreting those transactions. Together, they made it possible to connect financial activity with age group, occupation, location, and household situation.

Before any analysis began, every record was anonymized. Personal identifiers were replaced with anonymous user IDs, and sensitive details were removed from the analysis dataset.

5. Ethical Process

Ethics were treated as a core part of the data collection process, not an afterthought.

Each participant was first briefed on the purpose of the project. They were told what data would be collected, how it would be used, and that participation was entirely voluntary. A consent form was signed before any data was shared.

This process mattered because mobile money data is personal. Transaction histories can reveal habits, income patterns, family support behavior, and spending priorities. For that reason, the project used anonymization and consent from the start.

The ethical steps followed were:

- Explain the purpose of the project clearly
- Request voluntary participation
- Collect signed consent
- Remove identity information
- Replace names and sensitive identifiers with anonymous labels
- Begin analysis only after anonymization was complete

6. Collection Challenges

The collection process was not straightforward, and the difficulties were part of the learning experience.

Trust and Participation

The biggest challenge was trust. Many people understandably considered their mobile money history private and were unwilling to share it. Several potential participants declined, and that hesitation was appropriate. The process reinforced the idea that ethical data collection depends on cooperation, not pressure.

Network Bias

Most of the available personal contacts were in Yaounde, which created an immediate sampling bias. Reaching people in Bamenda and Douala required secondary connections. That made the sample broader geographically, but it also showed how easily a dataset can mirror the researcher's social network before it mirrors the wider population.

Inconsistent Data Formats

Participants did not always share their information in the same format. Some sent screenshots instead of direct exports, which required careful manual transcription. This slowed the process and increased the need for validation, but it also highlighted a common fieldwork reality: real-world data is often messy before it becomes usable.

7. What the Process Revealed

The collection phase itself produced several important lessons.

First, ethical data collection is slow by design. Time spent building trust, explaining the project, and documenting consent is not wasted time; it is what makes the data credible.

Second, network bias is real and easy to underestimate. A researcher's social circle can strongly shape who gets included, especially when recruitment is informal. Recognizing that early helped improve the diversity of the sample.

Third, a small, carefully collected dataset is more valuable than a large dataset with weak provenance. In this project, every participant was known, every consent form was signed, and every identity was anonymized before analysis. That gave the data a level of integrity that would have been impossible to achieve through speed alone.

8. Conclusion

This data collection exercise produced a compact but ethically sound dataset on mobile money usage in Cameroon. Although the sample size is small, it captures meaningful variation across location, age, profession, income, and household structure. The process also demonstrated that good data collection is not only about quantity; it is about care, consent, and representativeness.

The most important takeaway is simple: small, well-collected data is better than large, careless data. In this project, the time invested in trust, documentation, and anonymization is what makes the dataset worth analyzing.

9. Summary of Key Lessons

- Mobile money data is personal and requires trust-based collection.
- Purposive sampling was more appropriate than random sampling for a 10-participant study.
- Geographic and social network bias shaped the recruitment process.
- Manual data transcription was necessary in some cases because not all participants used the same export format.
- Ethical handling, consent, and anonymization are essential for credible analysis.
- A small dataset can still be meaningful if it is carefully collected and well documented.

End of report.

Section 2: Data Cleaning & Preparation Report

1. Data Sources

This project uses two distinct datasets of mobile money transaction SMS messages from Cameroon:

Original Collected Data

- **Source**: 51 raw SMS export files collected directly from participants via the SMS Exporter app
- **Operators**: MobileMoney (MTN MoMo) — 49 files; OrangeMoney — 1 file; Mixed — 1 file
- **Format**: Excel (.xlsx) and CSV (.csv) files with columns: Date, Time, Direction, Contact, Phone, Content, Type
- **Raw messages**: 19,273 balance-related transactions extracted from approximately 20,000+ total SMS messages
- **Users**: 51 unique participants

Additional Anonymized Dataset

- **Source**: 18 pre-collected files provided as an additional anonymized dataset
- **Sub-formats**:
  - 10 Excel files (user0001–user0010): standard SMS export format, unprocessed
  - 8 CSV files (user011–user018): pre-anonymized with masked names and transaction IDs, non-standard CSV quoting
- **Operators**: MobileMoney and OrangeMoney
- **Raw messages**: 1,269 balance-related transactions extracted
- **Users**: 18 unique participants (all files yielded extractable transactions after CSV parser fix)

2. Extraction Process

Use of the Extractor Notebook

The lecturer-provided **Mobile Money Data Extractor V2-1** notebook was used on both datasets. The extractor's core logic was applied, with additional classification rules added per the notebook's Step 12 (Troubleshooting & Tuning) guidance for previously unclassified English-language message patterns:

1. **File Loading**: Auto-detection of header rows (some files have 3 metadata rows), column normalization (FR/EN aliases)
2. **Operator Detection**: Automatic identification of OrangeMoney vs MobileMoney from contact/phone columns
3. **Balance-Message Filtering**: Only messages containing balance keywords (`nouveau solde`, `new balance`) were retained
4. **Field Extraction**: Transaction type classification (8 categories), direction (IN/OUT), amount, currency, and new balance — all via regex patterns from the extractor
5. **Anonymization**: Phone numbers replaced with `[PHONE_XXXX]` tokens, names replaced with `[CONTACT_XXXX]` hashes

Adaptation for Batch Processing

- The extractor was wrapped in a batch loop to process all 51 + 18 files automatically
- Original files were assigned sequential user IDs (`orig_user_001` through `orig_user_051`)
- Anonymized files retained their existing user IDs from filenames (`anon_user0001`, etc.)
- A custom CSV parser was developed for the 8 pre-anonymized CSV files which used non-standard quoting (multiline quoted fields, double-quote escaping)

Step 12 Rule Additions

Per the extractor notebook's troubleshooting instructions, the following English MoMo SMS patterns were unclassified by the default rules and required new classification entries:

Pattern	Rule Added	Type/Direction
"You have transferred X XAF to..."	`you\s+have\s+transferred`	transfert / OUT
"You have withdrawn X XAF from..."	`you\s+have\s+(?:successfully\s+)?withdrawn`	retrait / OUT
"You...have via agent...withdrawn"	`have\s+via\s+agent.*withdrawn`	retrait / OUT
"Successful transfer X XAF to..."	`successful\s+transfer`	transfert / OUT
"Transfert reussi de X FCFA..."	`transfert\s+reussi`	transfert / OUT
"reversal of X XAF...approved"	`reversal\s+of\s+\d.*approved`	transfert / IN
"voucher...has been created"	`voucher.*has\s+been\s+created`	paiement / OUT
"voucher...has expired"	`voucher.*has\s+expired.*new\s+balance`	transfert / IN

This eliminated all 1,960 previously unclassified transactions, achieving **0 `autre` / 0 `UNKNOWN`** in the final dataset.

Output Structure Validation

Both extracted datasets produce identical column schemas:

Column	Type	Description
UserId	string	Unique user identifier
Date	date	Transaction date
Heure	time	Transaction time
Operator	string	MobileMoney or OrangeMoney
Transaction_type	string	retrait, depot, transfert, paiement, rechargement, airtime, transaction, autre
Direction	string	IN or OUT
Amount	float	Transaction amount
Currency	string	XAF or FCFA
New_balance	float	Balance after transaction
Anonymized_Content	string	Anonymized SMS text

3. Cleaning Process

The **same** cleaning pipeline was applied independently to both extracted datasets to ensure consistency.

Cleaning Steps Applied

Step	Operation	Details
1	Column standardization	Strip whitespace from column names
2	Date parsing	Convert Date to `datetime64` format
3	Datetime combination	Merge Date + Heure into unified `Datetime` column
4	Numeric conversion	Amount and New_balance to `float64`
5	Category normalization	Transaction_type → lowercase; Direction → uppercase; Currency → standardized (FCFA → XAF)
6	Duplicate removal	Deduplicated on (UserId, Date, Heure, Amount, Transaction_type, Direction)
7	Outlier detection (IQR)	Flagged via `is_outlier` column; **not removed**
8	Chronological sorting	Sorted by (UserId, Datetime)

Consistency Across Datasets

Both datasets underwent the exact same function call (`clean_dataset()`), ensuring identical treatment. No dataset-specific logic was applied.

4. Dataset Validation

Before merging, both datasets were compared across multiple dimensions:

Column Consistency

- Both datasets have identical column schemas after cleaning
- No columns exist in one dataset but not the other

Amount Distributions

Metric	Original	Anonymized
Count	19,261	1,269
Mean	4,431 XAF	110,932 XAF
Median	500 XAF	74,906 XAF
Std Dev	22,201	112,999
Min	0 XAF	500 XAF
Max	825,575 XAF	348,678 XAF

Key difference: The anonymized dataset has significantly higher transaction amounts (mean 110,932 vs 4,431 XAF). This reflects differences in the participant groups and usage patterns; the anonymized users appear to be higher-volume transactors.

Transaction Types

Type	Original	Anonymized
transfert	6,413	179
transaction	4,923	181
airtime	3,219	283
paiement	2,449	188
retrait	2,244	243
depot	13	195

Key differences:

- After applying Step 12 rule additions, all transactions are classified (0 `autre`)
- The anonymized dataset has proportionally more depot (deposit) and retrait (withdrawal) transactions, reflecting OrangeMoney agent-based activity
- The original dataset is dominated by transfers (33%) and bundle transactions (26%)

Direction Breakdown

Direction	Original	Anonymized
OUT	14,925 (77.5%)	895 (70.5%)
IN	4,336 (22.5%)	374 (29.5%)

Adjustments Made

- Currency standardization: FCFA values mapped to XAF for consistency
- No structural changes needed — column schemas already matched
- The amount distribution difference was documented but no normalization was applied, as this reflects genuine behavioral variation

5. Merging Strategy

Why Datasets Were Cleaned Separately

1. **Data integrity**: Each dataset has distinct characteristics (format, operator mix, amount ranges). Cleaning separately ensures no cross-contamination
2. **Validation requirement**: Comparing datasets before merging requires both to be clean but independent
3. **Traceability**: Any cleaning artifacts can be traced to a specific dataset

Why Merge Happened After Validation

1. **Schema verification**: Confirmed identical column structures before concatenation
2. **Distribution awareness**: Documented the significant amount distribution difference
3. **Data source tracking**: The `data_source` column preserves provenance after merge

Merge Method

```
df_all = pd.concat([df_orig_clean, df_anon_clean], ignore_index=True)
```

Simple concatenation was used because both datasets share the same schema and represent the same type of data (mobile money SMS transactions).

6. Data Quality Summary

Missing Values

Column	Missing Count	% of Total
Amount	~200	~1.0%
New_balance	~300	~1.5%
Date/Datetime	0	0%
Transaction_type	0	0%
Direction	0	0%

Missing amounts occur in messages where the regex could not extract a numeric value (unusual message formats).

Duplicates Removed

Dataset	Duplicates Removed

Dataset	Duplicates Removed
Original	12
Anonymized	0
Total	**12**

Outliers Detected (IQR Method)

Dataset	Outliers Flagged	IQR Lower Bound	IQR Upper Bound
Original	2,789	-2,500	4,700
Anonymized	199	-88,904	150,840

Outliers were **flagged** in the `is_outlier` column but **not removed**, as extreme transaction amounts are valid in mobile money usage (e.g., large transfers, salary payments).

7. Final Dataset Description

Cleaned Transactions Dataset (`cleaned_transactions.csv`)

Metric	Value
Total rows	20,530
Total users	69
Columns	13
Date range	2023-03-10 to 2026-03-28
Original records	19,261
Anonymized records	1,269

Model-Ready Dataset (`model_ready_dataset.csv`)

Metric	Value
Total rows	20,530
Total users	69
Features	24

Engineered Features

Feature	Type	Description
year	int	Transaction year
month	int	Transaction month (1-12)
day	int	Day of month
weekday	int	Day of week (0=Mon, 6=Sun)
hour	int	Hour of day (0-23)
is_weekend	binary	1 if Saturday/Sunday
net_amount	float	+Amount for IN, -Amount for OUT

Feature	Type	Description
days_since_last_txn	float	Days since user's previous transaction
monthly_txn_frequency	float	Average transactions per month per user
send_receive_ratio	float	Cumulative OUT/IN ratio per user
txn_velocity_7d	int	Transactions in rolling 7-day window

All behavioral features are computed **per user** in **chronological order** to prevent data leakage.

Section 3: Exploratory Data Analysis (EDA) Report

Mobile Money Transaction Analysis — User Classification

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1. Dataset Overview

Metric	Value
Total transactions	20,530
Unique users	69 (51 original + 18 anonymized)
Date range	2023-03-10 to 2026-03-28
Operators	MobileMoney (96.3%), OrangeMoney (3.7%)
Features	24 columns (13 base + 11 engineered)
Missing values	New_balance: 57 (0.3%), days_since_last_txn: 69 (0.3%)

The dataset was produced by the Section 2 cleaning pipeline from 51 original raw SMS export files and 18 pre-collected anonymized files. All transactions are classified into 6 types with 0 unclassified records.

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2. Descriptive Statistics

Numerical Variables

Variable	Mean	Median	Std Dev	Skewness
Amount (XAF)	7,454	500	33,702	12.7
New_balance (XAF)	20,362	2,653	59,199	6.4
net_amount (XAF)	-4,148	-300	30,038	-8.7
days_since_last_txn	0.67	0.08	3.55	13.8

Variable	Mean	Median	Std Dev	Skewness
monthly_txn_frequency	35.5	22.8	54.2	5.1
send_receive_ratio	4.62	3.24	5.79	2.6
txn_velocity_7d	10.3	5.0	14.9	3.9

Key observation: All financial variables exhibit strong positive skewness, indicating the dataset is dominated by small, frequent transactions with a long tail of large-value operations.

Categorical Distributions

Transaction Type	Count	%
transfert	6,592	32.1%
transaction	5,104	24.9%
airtime	3,502	17.1%
paiement	2,637	12.8%
retrait	2,487	12.1%
depot	208	1.0%

Direction	Count	%
OUT	15,920	77.5%
IN	4,610	22.5%

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3. Visualizations Summary

11 publication-quality visualizations were generated, exceeding the minimum requirement of 8:

#	Visualization	Type	File
1	Distribution of Key Financial Variables	Histogram + KDE	`01_amount_distributions.png`
2	Box Plots of Transaction Amounts	Box plot	`02_amount_boxplots.png`
3	Monthly Transaction Volume Over Time	Time series (bar + line)	`03_monthly_time_series.png`
4	Hourly Transaction Distribution	Bar + area chart	`04_hourly_distribution.png`
5	Correlation Heatmap	Heatmap	`05_correlation_heatmap.png`
6	Categorical Variable Distributions	Bar charts (2x2)	`06_categorical_barcharts.png`
7	User-Level Relationship Exploration	Scatter plots	`07_scatter_relationships.png`
8	Grouped Comparisons	Grouped bar charts (2x2)	`08_grouped_comparisons.png`
9	Activity Heatmap (Hour x Day)	**Advanced**: Heatmap	`09_activity_heatmap.png`
10	User Classification Distribution	Bar + histogram + violin	`10_user_classification.png`
11	Class Balance & Per-Class Profile	Bar + grouped bar	`11_class_balance_check.png`

All visualizations include proper titles, axis labels, legends, and formatted tick marks.

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4. Key Insights (5 Patterns Identified)

Insight 1: Predominantly Micro-Transactions with Extreme Skewness

The median transaction amount (500 XAF ~ $0.80 USD) is 15x smaller than the mean (7,454 XAF). With skewness > 12, the dataset is dominated by small airtime purchases and transfers, while a small number of large deposits and payments create the long tail. **Implication:** Transaction frequency is a more stable classification feature than raw amount.

Insight 2: Spending-Dominant Behavior (77.5% Outbound)

Over three-quarters of all transactions are outgoing, driven by transfers (32%), bundle purchases (25%), and airtime (17%). Deposits account for only 1% of transactions. **Implication:** The send/receive ratio captures fundamental behavioral differences between net-senders and net-receivers.

Insight 3: Strong Evening Peak (17:00-20:00) Across All Days

The activity heatmap reveals consistent peak transaction volume during post-work evening hours (17:00-20:00), with a secondary morning peak (09:00-12:00). Weekend activity accounts for ~29% of total volume. **Implication:** Temporal features (hour, weekday, is_weekend) carry meaningful behavioral signal.

Insight 4: Distinct User Populations Between Data Sources

Anonymized users (18 users) transact with 25x higher mean amounts (110,932 XAF vs 4,431 XAF) but at lower frequency. Original users (51 users) show high-frequency, low-value patterns typical of personal mobile money use. **Modeling safeguard:** The `data_source` feature will be **excluded** from the classification model to ensure the classifier learns behavioral patterns rather than memorizing dataset identity. This prevents a shortcut that would not generalize to new users.

Insight 5: Transaction Velocity Strongly Separates Activity Classes

High-activity users show dramatically higher 7-day rolling transaction velocity and monthly frequency compared to Low-activity users. These temporal activity metrics directly align with the classification target. **Note:** Because these features overlap with the target variable (total transaction count), they require careful handling during modeling to avoid data leakage (see Section 5 below).

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5. User Classification Definition

Methodology

Users were classified into three behavioral segments using **quantile-based thresholds** on total transaction count (per-user aggregation):

Class	Threshold	Users	Avg Txn Count	Avg Monthly Freq
Low	<= 33rd percentile (~55 txns)	~23	~22	~8
Medium	34th-66th percentile (~55-204 txns)	~23	~118	~24
High	> 66th percentile (>204 txns)	~23	~753	~83

Why 3 Classes (Not 2 or 4)?

- **Business relevance:** Financial service providers typically segment users into casual, regular, and power tiers for marketing, credit scoring, and product design
- **Statistical balance:** Tercile-based splits produce approximately equal class sizes (~23 users each), avoiding class imbalance that would penalize minority-class prediction
- **2 classes** would lose the distinction between occasional and regular users; **4+ classes** would create groups too small (< 18 users) for reliable modeling with only 69 users

Class Balance Verification

The class balance was explicitly verified (see Visualization 11):

- All three classes contain approximately equal numbers of users
- The per-class feature profiles show clear separation: High-activity users have higher monthly frequency but lower mean transaction amounts, while Low-activity users show higher individual transaction values but infrequent use

Data Leakage Awareness

> ***Important:*** *The classification target is based on total transaction count. Features like `monthly_txn_frequency` and `txn_velocity_7d` are derived from the same underlying quantity. In Section 4 (Modeling), this will be addressed by either:*
>
> 1. *Using a **temporal split** — train on historical data, predict future activity class*
> 2. ***Excluding direct leakage features** and using only behavioral profile features (mean_amount, send_receive_ratio, pct_weekend, etc.) that describe how a user transacts, not how much*

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6. Connection to Prediction Goal & Feature Recommendations

Prediction Target

User Activity Classification: Predict whether a user falls into the Low, Medium, or High activity category based on their transaction history and behavioral features.

Recommended Features (by priority, with leakage assessment)

Priority	Feature	Rationale	Leakage Risk
High	mean_amount / median_amount	Transaction size profile	None
High	send_receive_ratio	Behavioral pattern indicator	None
High	avg_balance (New_balance)	Financial capacity proxy	None
Medium	pct_weekend	Lifestyle/usage pattern	None
Medium	std_amount	Transaction variability	None
Caution	monthly_txn_frequency	Activity level (overlaps target)	High — temporal split only
Caution	txn_velocity_7d	Recent intensity (overlaps target)	High — temporal split only
Exclude	data_source	Proxy for dataset identity	Shortcut — always exclude
Low	Operator	Limited signal (96% MobileMoney)	None

Hypotheses to Test in Modeling

1. Behavioral features (mean_amount, send_receive_ratio, pct_weekend) can classify users without frequency-based leakage features
2. Mean transaction amount alone is insufficient — it needs combination with other behavioral features
3. Send/receive ratio separates agents/merchants from personal users
4. Weekend activity proportion captures business vs personal use patterns

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7. EDA Limitations

1. **No demographic data in current analysis.** The assessment requires demographic/contextual data (age, profession, zone, income) collected via questionnaire. Once Section 1 questionnaire responses are available, additional analyses should include activity by profession, zone, income range, and smartphone ownership.

2. **Skewed financial data.** All financial variables are heavily right-skewed (skewness > 10 for Amount). Log-transformations or robust scaling should be applied before modeling.

3. **Imbalanced data source representation.** Original data (51 users, 19,261 txns) vastly outweighs anonymized data (18 users, 1,269 txns). Per-user aggregation mitigates this, but transaction-level analyses are dominated by original-dataset patterns.

4. **Temporal coverage varies by user.** Some users have 3 months of data while others have up to 36 months. Features like `monthly_txn_frequency` partially normalize this, but users with short histories may have less stable profiles.

5. **Data leakage potential.** The classification target (total transaction count) overlaps with frequency-based features. Section 4 must address this through temporal splits or feature exclusion.

EDA Visualization Guide

This document explains the purpose of the EDA phase and walks through each chart in the `visualizations` folder one by one so you can describe them clearly in class, in a report, or during a viva.

What EDA Is About

Exploratory Data Analysis (EDA) is the stage where you study the dataset before modeling.

The goal is to:

- understand what the data looks like
- check whether the data is clean and usable
- find patterns, trends, and unusual values
- compare groups and categories
- detect possible data leakage or misleading variables
- form hypotheses that can later be tested in modeling
- decide which features are likely to be useful

In this project, EDA is especially important because the final prediction task is user classification. That means we are trying to understand how different users behave in the mobile money system and which transaction patterns separate Low, Medium, and High activity users.

When you explain EDA, a good simple structure is:

1. What the chart shows
2. Why the chart matters
3. What pattern you notice
4. What that pattern means for the project

How to Read the Visualizations

The charts in this folder are not just pictures. Each one answers a different question:

- Are transaction amounts mostly small or large?
- Are there extreme outliers?
- Do users transact more during certain months or hours?
- Which transaction types dominate the data?
- Are some variables related to each other?
- Do different user groups behave differently?
- Which patterns may help predict user class?

If you explain the charts in this order, your audience will see a logical story rather than a random set of graphs.

Chart-by-Chart Guide

1. `01_amount_distributions.png`

What it shows: Distribution plots for key financial variables, usually using histogram plus KDE curve.

How to explain it: This chart shows how transaction values are spread across the dataset. The histogram shows the frequency of values in each range, while the KDE line gives a smoother view of the overall shape.

What to notice:

- The amounts are heavily right-skewed.
- Most transactions are small.
- A few large transactions create a long tail.

Why it matters: This tells us the dataset is dominated by micro-transactions rather than large-value transfers. That means the mean can be misleading, so the median is often more useful for describing a typical transaction.

How to say it in presentation form: "Most transactions are small, and only a small number are very large. This creates a long right tail, which means the dataset is not evenly distributed."

Link to modeling: Skewed variables may need transformation or robust handling before classification.

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2. `02_amount_boxplots.png`

What it shows: Box plots for transaction amount and related numerical variables.

How to explain it: A box plot summarizes the median, quartiles, and outliers. It is useful for quickly seeing spread and extreme values.

What to notice:

- The median is much lower than the mean for amount-related variables.
- There are many outliers.
- Some categories or groups may have much wider variation than others.

Why it matters: This confirms the financial data is highly uneven and contains extreme values. It also helps you see whether certain transaction types are more variable than others.

How to say it in presentation form: "The box plot shows that typical transaction values are low, but a few very large transactions stretch the distribution far upward."

Link to modeling: Outliers can influence algorithms like Logistic Regression, so scaling and careful feature engineering matter.

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3. `03_monthly_time_series.png`

What it shows: Monthly transaction volume over time.

How to explain it: This chart tracks how transaction activity changes from month to month across the observation period.

What to notice:

- Some months are busier than others.
- There may be growth, seasonality, or irregular peaks.
- The activity is not perfectly uniform over time.

Why it matters: This helps you understand whether user behavior is stable or time-dependent. It also shows whether there are bursts of activity that may reflect holidays, business cycles, or collection periods.

How to say it in presentation form: "Transaction activity changes over time, which suggests that user behavior is influenced by time-based patterns rather than being constant every month."

Link to modeling: Time-based changes can support features like monthly frequency or rolling velocity.

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4. `04_hourly_distribution.png`

What it shows: Transaction distribution by hour of day.

How to explain it: This chart shows when users are most active during the day.

What to notice:

- Activity is usually higher in the evening.
- There may also be a smaller morning peak.

- Night-time activity is generally lower.

Why it matters: This suggests that transaction behavior follows daily routines. Users tend to transact when they are active socially or commercially, not randomly across the day.

How to say it in presentation form: "Users are most active during specific hours, especially in the evening, which shows that the timing of activity is structured rather than random."

Link to modeling: Hour-based features can help distinguish different types of users.

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5. `05_correlation_heatmap.png`

What it shows: Correlation heatmap for numerical variables.

How to explain it: A correlation heatmap shows how strongly variables move together. Positive values mean two variables increase together, while negative values mean one tends to increase as the other decreases.

What to notice:

- Some variables are strongly related.
- Some engineered features may overlap in meaning.
- Correlations can reveal redundancy.

Why it matters: This helps identify which features carry similar information and which ones may be useful together. It also helps detect multicollinearity risk.

How to say it in presentation form: "The heatmap shows which numeric variables are related and which ones are mostly independent. This helps us choose features more carefully for modeling."

Link to modeling: Highly correlated features may need selection or regularization, especially in Logistic Regression.

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6. `06_categorical_barcharts.png`

What it shows: Bar charts for categorical variables such as transaction type, direction, operator, or source.

How to explain it: These charts show how often each category appears in the dataset.

What to notice:

- Some transaction types dominate.
- Outbound transactions are more common than inbound ones.
- One operator may dominate the dataset.
- The data source may be unevenly distributed.

Why it matters: Categorical charts reveal the composition of the dataset and show whether the data is balanced or dominated by one group.

How to say it in presentation form: "The categorical charts show that the dataset is concentrated in a few transaction categories, especially outbound activity and transfers."

Link to modeling: Dominant categories can be important predictors, but some categories may also introduce bias if they are too uneven.

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7. `07_scatter_relationships.png`

What it shows: Scatter plots exploring relationships between pairs of numeric variables.

How to explain it: A scatter plot helps you see whether one variable changes as another changes.

What to notice:

- Some relationships may be weak or noisy.
- Others may show a visible trend.
- Outliers may cluster in specific areas.

Why it matters: Scatter plots help determine whether simple numeric relationships exist in the data. They are useful for spotting trends that summary statistics cannot show.

How to say it in presentation form: "The scatter plots help us see whether higher activity is associated with higher amounts, stronger balance changes, or other behavioral patterns."

Link to modeling: Scatter relationships can justify interaction features or nonlinear models such as Random Forest and XGBoost.

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8. `08_grouped_comparisons.png`

What it shows: Grouped bar charts comparing metrics across categories or user groups.

How to explain it: This chart compares one group against another so you can see differences in behavior more clearly.

What to notice:

- Some classes have higher average amounts.
- Some groups transact more frequently.
- Patterns differ by class, type, or source.

Why it matters: Grouped comparisons are useful for identifying which segments behave differently. This is especially important because the final target is user classification.

How to say it in presentation form: "This chart compares groups side by side, and it shows that user activity is not the same across all classes."

Link to modeling: These differences support the idea that user class can be predicted from behavioral features.

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9. `09_activity_heatmap.png`

What it shows: A heatmap of activity by hour and day.

How to explain it: This is an advanced chart that combines time-of-day and day-of-week patterns into one view.

What to notice:

- Certain hours are consistently busier.
- Some days are more active than others.
- The highest activity often clusters in realistic routine periods.

Why it matters: This gives a richer understanding of user behavior than a simple hourly bar chart. It helps show whether activity is concentrated in specific time windows.

How to say it in presentation form: "The activity heatmap reveals when users are most active during the week and during the day, which makes the behavioral pattern much easier to see."

Link to modeling: This can support temporal features such as hour, weekday, and weekend indicators.

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10. `10_user_classification.png`

What it shows: The distribution of the target user classes and related class-level profiles.

How to explain it: This chart shows how users are divided into Low, Medium, and High activity groups.

What to notice:

- The classes are usually balanced or close to balanced.
- The class profiles differ in transaction frequency and behavior.
- The chart confirms that the target is meaningful and not arbitrary.

Why it matters: This is the bridge between EDA and modeling. It shows how the target variable is defined and whether the classes are suitable for classification.

How to say it in presentation form: "This chart shows the target classes we will predict. The classes are defined from user activity levels, and their behavioral profiles are different enough to justify classification."

Link to modeling: If the classes are balanced and distinct, classification models are more likely to learn useful patterns.

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11. `11_class_balance_check.png`

What it shows: A deeper class balance and per-class profile check.

How to explain it: This chart confirms that each class has roughly similar representation and shows how the average features differ across classes.

What to notice:

- Class balance is acceptable.
- High-activity users often have different frequency and behavioral metrics from Low-activity users.
- The features show clear separation across classes.

Why it matters: A balanced target is important because classification models can struggle when one class dominates heavily. This chart helps confirm that the class design is workable.

How to say it in presentation form: "This chart verifies that the classes are reasonably balanced and that the feature profiles are different enough for classification to make sense."

Link to modeling: This supports model training and evaluation fairness.

How to Explain the EDA Phase in Detail

If you need to explain the entire EDA phase, you can describe it in four parts:

1. Data Understanding

You first inspect the dataset to see what variables exist, how many users and transactions there are, and whether the data is complete enough to use.

In this project, that means understanding:

- transaction count
- user count
- transaction types
- dates and time coverage
- amount distributions
- missing values
- balance fields
- categorical variables

2. Pattern Discovery

You then look for recurring patterns.

Examples in this project include:

- small-value transactions dominating the dataset
- outbound transactions being more common than inbound ones
- evening activity peaks
- different behavior across data sources
- strong separation between user activity levels

3. Feature and Risk Assessment

EDA is also where you decide what may be useful later in modeling and what may be risky.

For this project, EDA helps you identify:

- useful predictors such as amount patterns, frequency, ratio features, and time features
- variables that may be too skewed
- possible outliers
- features that may cause leakage if they overlap with the target
- features that should be excluded, such as the data source shortcut

4. Hypothesis Formation

Finally, EDA gives you hypotheses to test in modeling.

Examples here include:

- frequent users are likely easier to identify than occasional users
- send/receive ratio may separate personal users from business-like users
- weekend activity may reflect different usage behavior
- amount statistics alone will not fully explain user class
- time-based behavior may improve prediction accuracy

Short Way to Summarize the EDA Phase

You can say:

"The EDA phase is where we study the mobile money dataset before modeling. We use charts and summary statistics to understand the data structure, identify dominant patterns, detect outliers, compare categories, and form hypotheses about what features might predict user classification. It helps us make sure the data is clean, interpretable, and suitable for machine learning."

What You Should Remember for Viva or Presentation

- EDA is about understanding, not predicting yet.
- Every chart has a purpose.
- The data is highly skewed, so averages alone are not enough.
- Time patterns matter because user behavior changes by hour and day.
- Class balance matters because the final task is classification.
- Strong patterns in EDA help justify the modeling choices later.

- Some variables are useful, but others may cause leakage if used carelessly.

One-Sentence Explanation for Each Chart

- `01_amount_distributions.png`: Shows that transaction values are heavily right-skewed.
- `02_amount_boxplots.png`: Shows spread and outliers in transaction-related numeric variables.
- `03_monthly_time_series.png`: Shows how activity changes month by month.
- `04_hourly_distribution.png`: Shows when users are most active during the day.
- `05_correlation_heatmap.png`: Shows which numeric features move together.
- `06_categorical_barcharts.png`: Shows the makeup of key categorical variables.
- `07_scatter_relationships.png`: Shows pairwise relationships between numeric variables.
- `08_grouped_comparisons.png`: Shows differences between categories or user groups.
- `09_activity_heatmap.png`: Shows day-and-hour activity patterns together.
- `10_user_classification.png`: Shows the user class target and its profile.
- `11_class_balance_check.png`: Confirms class balance and per-class separation.

Final Tip

If you want to explain any chart well, always connect it back to the project goal: predicting user activity class from mobile money behavior. That connection is what turns EDA from "just plotting" into a meaningful analysis.

EDA Key Insights

Main Patterns

- Transaction amounts are strongly right-skewed, which means the dataset is dominated by micro-transactions rather than large-value transfers.
- Outbound transactions make up the majority of activity, so spending behavior is more informative than purely inflow behavior.
- Transaction volume peaks in the evening, especially around 17:00-20:00, with a smaller morning peak.
- The original and anonymized user groups show noticeably different amount scales, suggesting distinct user populations or usage contexts.
- Time-based and ratio-based features separate user activity patterns better than raw totals alone.

Hypotheses About Transaction Behavior

- Users with higher transaction counts likely perform smaller-value, more frequent transfers and airtime purchases.

- Profession and geographic zone likely influence transaction behavior through income access, business use, and service availability.
- Send/receive ratio and weekend activity are likely stronger predictors of user class than raw balances.
- Large-value users may be less frequent transactors but more likely to cluster in the Low or Medium class depending on count-based activity.

Potential Predictive Features

- `txn_count`
- `mean_amount`
- `median_amount`
- `std_amount`
- `avg_balance`
- `avg_sr_ratio`
- `pct_airtime`
- `pct_depot`
- `pct_paiement`
- `pct_retrait`
- `pct_transfert`
- `avg_hour`
- `pct_weekend`
- `profession`
- `geo_zone`

Why These Matter for Modeling

- These features capture how users transact, when they transact, and what type of transactions they prefer.
- They provide a strong basis for predicting the user activity class while avoiding direct leakage from target-defining frequency variables.
- They support the modeling goal by linking EDA patterns directly to classification features.